

2 Asymptotic Analysis

2.1 Proofs

Prove the following for big O.

$$n^2 + 2n + \lg n = O(n^2) \quad (1)$$

$$\leq n^2 + 2n + \lg n \quad (2)$$

$$\leq n^2 + 2n^2 + \lg n \cdot n \quad (3)$$

$$\leq n^2 + 2n^2 + n^2 \quad (4)$$

$$= 4n^2 \quad (5)$$

So $c = 4$ and $n_0 = 1$.

2.2 Proof II

Prove the following for big Ω

$$2n^3 - 3n^2 + 2n = \Omega(n^3) \quad (6)$$

$$= 2n^3 - 3n^2 + 2n \quad (7)$$

$$\geq 2n^3 - 3n^2 \quad (8)$$

$$\geq 2n^3 - n^3, \quad n \geq 3 \quad (9)$$

$$= n^3 \quad (10)$$

So $c = 1$, $n_0 = 3$.

2.3 Proof III

Upper bound.

$$\frac{n^2}{2} - \frac{n}{2} = \Theta(n^2) \quad (11)$$

$$= \frac{n^2}{2} - \frac{n}{2} \quad (12)$$

$$\leq \frac{n^2}{2} - \frac{n}{2} \quad (13)$$

$$= \leq \frac{n^2}{2}, c_2 = \frac{1}{2}, n \geq 0. \quad (14)$$

Lower bound.

$$\frac{n^2}{2} - \frac{n}{2} \quad (15)$$

$$\geq \frac{n^2}{2} - \frac{n}{2} * \frac{n}{2} \quad (16)$$

$$= \frac{1}{4}n^2 \quad (17)$$

$$c_1 = \frac{1}{4} \text{ and } n_0 = 2$$

2.4 Growth Classes Proof

Show $n^3 - 10n^2 \neq O(n^2)$

$$0 \leq n^3 - 10n^2 \leq cn^2 \quad (18)$$

$$n^3 \leq 10n^2 + cn^2 \quad (19)$$

$$n^3 \leq (10 + c)n^2 \quad (20)$$

$$n \leq 10 + C \quad (21)$$

This is a contradiction.

2.5 Growth Classes Example II

Show $2^n = o(3^n)$

$$2^n = o(3^n) \quad (22)$$

$$\lim_{n \rightarrow \infty} \frac{2^n}{3^n} = \lim_{n \rightarrow \infty} \frac{2^n}{3} \quad (23)$$

$$= 0 \quad (24)$$